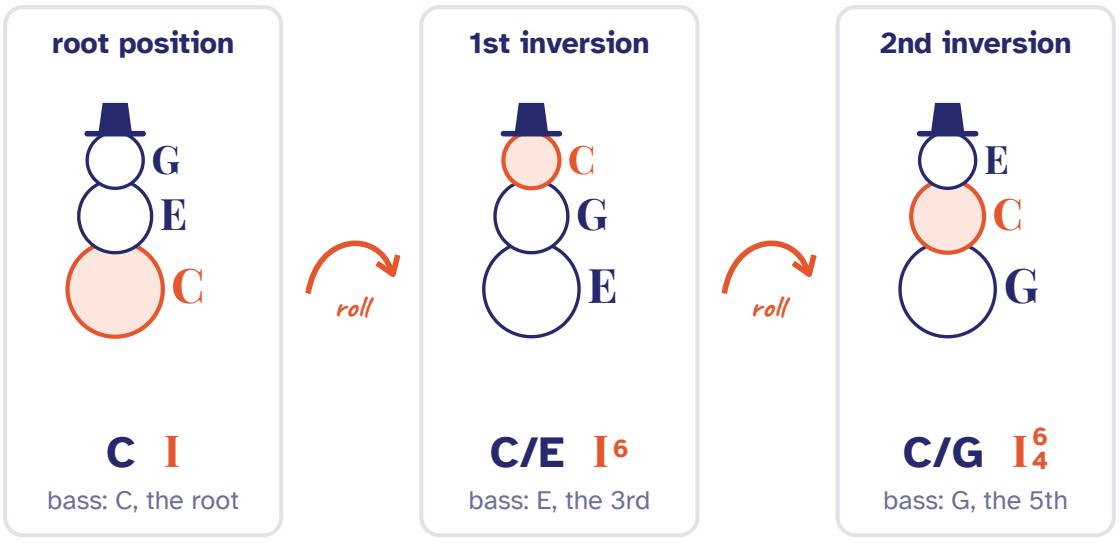
PART IV · CHORDS · 4

# Inversions

Same three notes, a different one at the bottom. The chord keeps its name; only the bass changes. The red ball is the root, C: watch it travel.

**THE SECRET**

**Move the bottom note to the top: same chord, new bass.**

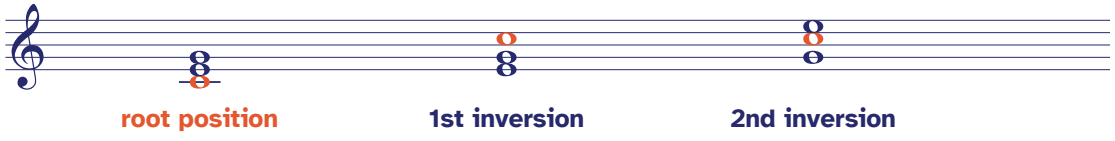



1. Start with the snowman **C E G**: root at the bottom.
2. Roll the bottom ball to the top: **E G C**. That is the 1st inversion.
3. Roll again: **G C E**, the 2nd inversion. Still **C major**.

## Two ways to write it

| | ROOT POSITION | 1ST INVERSION | 2ND INVERSION |
|---|---|---|---|
| **Chord symbol** pop, jazz | C | C/E | C/G |
| **Harmony class** numerals | I | $I^6$ | $I^6_4$ |
| **Also written** | Ia, 5/3 | Ib, 6/3 | Ic, 6/4 |
| **First met** | Level ½ | Level 3 | Level 5 |

The little numbers count up from the bass. E G C: C is a **6th** above E, so $I^6$. G C E: E is a 6th and C a **4th** above G, so $I^6_4$. A slash chord just names the bass: C/E (more on p. 108).




The same three chords on the staff; the red note is the root, C. To find the root, restack the notes every other letter (like C E G): then the bottom ball is the root.

PART IV · CHORDS · 4 · CONTINUED

## Why inversions?

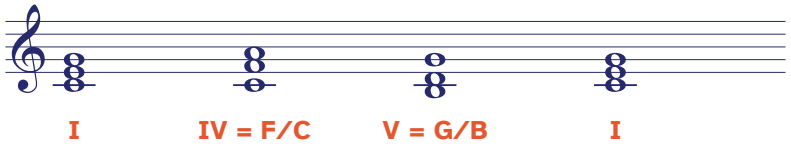
They let your hand **move as little as possible**: keep any note two chords share, and move the others to the nearest note of the next chord. Left-hand accompaniments and hymn parts are written this way.

## Chords of C, every position

| CHORD | ROOT POSITION | 1ST INVERSION | 2ND INVERSION |
|---|---|---|---|
| **I** (C) | C E G | E G C | G C E |
| **IV** (F) | F A C | A C F | C F A |
| **V** (G) | G B D | B D G | D G B |
| **vi** (Am) | A C E | C E A | E A C |

EXAMPLE 1

### Play I, IV, V, I smoothly




1. **I:** C E G. **IV:** keep C, move E up to F and G up to A: C F A.
2. **V:** no shared note. C steps down to B, F down to D, A down to G: B D G.
3. **I again:** B up to C, D up to E, G stays: C E G.

IV is F/C and V is G/B: **inversions** keep the hand in one place.

GAME

### Which position?

| 1 | 2 | 3 |
|---|---|---|
| C E G | E G C | G C E |

Same three notes. Look at the bottom one: root, 3rd or 5th?

Answer: 1 root position · 2 first inversion (C/E) · 3 second inversion (C/G).

TRY IT · 10 SECONDS




### Climb the chord.

Play C E G, then E G C, then G C E, then C E G an octave up. Same snowman, turning over as it climbs.

PART IV · CHORDS · 4 · CONTINUED

EXAMPLE 2

### Write F major in first inversion




1. **F major:** F A C.
2. **First inversion:** the 3rd, A, goes to the bottom.
3. **Stack it:** A C F.

First inversion of F: **A C F**, written **F/A** or **IV⁶** in C.

EXAMPLE 3

### Find the second inversion of G
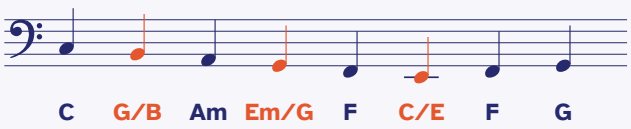



1. **G major:** G B D.
2. **Second inversion:** the 5th, D, goes to the bottom.
3. **Stack it:** D G B.

Second inversion of G: **D G B**, written **G/D** or **V⁶₄** in C.

YOU KNOW THIS

### Pachelbel's Canon, the bass (here in C)




In the popular version the bass walks **straight down the scale**: C B A G F E. It can, because some chords are **inversions**: G/B, Em/G and C/E put a different note in the bass.

**Try it.** Play the bass slowly and hum the famous violin tune over it.

THINK OF IT The same three friends in a photo, swapping places: still the same group (the same chord), but whoever stands at the front (the bass) changes the picture.

TRAP

- × Naming a chord from its bass note: in an inversion the bass is not the root. Restack it in 3rds first.
- × Thinking an inversion is a new chord: E G C is still C major.

SEE ALSO

Chords and triads p. 87 · Chords in a key p. 93 · Cadences p. 116

PRACTISE IT

Workbook Harmony 4 · Online: Library › Chords